

An AI Agentic Exploration of the Genetic Basis of 5'UTR Repeat Expansion-Induced Neuromuscular Degeneration.

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Abstract:

Thousands of trinucleotide repeats reside in the 5'UTR region of human genes, and nine of those are known to cause diseases when the number of repeats is expanded. Currently, it is unclear what distinguishes these pathogenic repeats from non-pathogenic ones. Using the AI agent Biomni and the AlphaGenome inference model, we explored the utility of LLM-based agents to predict pathomechanistic outcomes. The agent was informed of seven of the nine pathogenic repeats and asked to first identify what distinguishes them from the background and then use the findings to predict additional pathogenic repeats. We found that with the latest reasoning models, the agent identified the other two pathogenic repeats among the top 20 predictions ($p < 0.001$). We report our strategy, putative mechanisms, and candidate pathogenic repeats identified by this AI-agentic expedition.

Background

Short tandem repeats (STRs) are DNA elements composed of consecutive 1-6 bp motifs. The estimate of such repeats in the human genome ranges from less than 1 million to over 10 million, depending on the methodology used to identify them^{1,2}. A recent cataloging effort employing both reference-based and cohort-based approaches identified about 4.8 million such repeats².

STRs are inherently unstable and subject to either expansion or contraction. Over sixty STRs are currently known to cause disease when expanded^{3,4}, including Fragile X syndrome (FRAXA), Huntington's Disease (HD), twelve spinocerebellar ataxias (SCAs), *C9orf72*-linked amyotrophic lateral sclerosis/frontotemporal dementia (C9-ALS/FTD), and myotonic dystrophy (DM). STR-based pathomechanisms, which have been the subject of extensive research, include host gene haploinsufficiency, RNA gain-of-function⁵, and repeat-associated non-AUG (RAN) translation⁶.

It is currently unclear why certain STRs are susceptible to pathogenic expansions while the majority are not. A characteristic feature of disease-associated STRs is that GC-rich expansions form DNA structures that facilitate further expansion, although disease-linked AT-rich and GAA expansions also occur. However, it remains unclear which STR structural and/or sequence-context features render them susceptible to pathogenic expansions. The recent development of AI and LLM-based agents has provided a new opportunity to address this problem in a systematic and unbiased manner. In this brief communication, we explored the utility of AI agents in identifying disease-associated STRs.

Results

Datasets for 5'UTR TNRs.

After surveying various STR datasets, we decided to use the one cataloged by Weisburd et al.², which we refer to as the B (Broad Institute)-catalog. In addition to referencing the STRs to the hg38 human genome, this catalog also includes UCSC 36-mer mappability data flanking the repeats, and calculated population variation for each repeat locus based on Illumina 174K and T2T assembly data. A unique feature of the B-catalog is its organization of TNRs based on “canonical motif,” which considers all cyclic shifts and selects the one that’s alphabetically first as the representative motif². For instance, motifs CCG, CGC, GCC, CGG, GCG, GGC all share the canonical motif CCG.

This dataset contains the 63 repeats known to cause various degenerative neuromuscular diseases when their copy numbers are expanded. Of those, 10 are located in the 5'UTR (5' untranslated region) of genes, of which 9 are trinucleotide repeats (TNRs) (Figure 1A). Notably, canonical motifs for the 5'UTR pathogenic repeats are either CCG or AGC (Figure 1A). This prompts us to narrow our focus to these two types of 5'UTR repeats. There are a total of 5,446 canonical CCG and 1,204 canonical AGC 5'UTR repeats in the B-catalog (v1.0) (Figure 1B). These repeats (a total of 6,650) and associated information, including repeat length, mappability, and population variations, etc., were extracted from the B-catalog. The columns containing pathogenicity information were removed before feeding them into the AI agent (Supplementary file 1) (Figure 1C).

To infer the genomic consequence of repeat extension at each 5'UTR TNR, we used the AlphaGenome⁷ model to predict the impact of 2x, 5x, and 20x expansion of these repeats (Figure 1C). We selected all modalities available in AlphaGenome, including gene expression, RNA splicing, chromatin accessibility, histone marks, and 3D genome contacts. The AlphaGenome inference output was flattened to .tsv format, resulting in 3 genomic impact files (2x, 5x, 20x) of about 30-40 GB each (Supplementary files 2-4).

Comparison of LLM-based agentic runs in identifying the pathogenic repeats.

To evaluate the success of the agentic runs, we developed a leave-two-out strategy (Figure 1A). We reasoned that the soundness of the mechanistic hypotheses can be evaluated by whether the agentic run, informed by the discovered hypotheses, can identify the two pathogenic repeats that were withheld from the agent. Our earlier testing indicated that for this kind of open-ended hypothesis-generation task, the decoding temperature has a significant impact on the outcome. Thus, we chose to run each LLM at temperatures of 0.5, 0.7, and 0.9 (Figure 1D).

Of the 24 agentic runs conducted in December 2025, 14 completed the analysis and generated a list of predictions based on the identified hypotheses. Interestingly, five agentic runs (Opus 4.5 T=0.5, 0.7, 0.9; GPT5 T=0.5; GPT5.2 T=0.9) were able to identify both keys – the two pathogenic 5'UTR TNRs withheld from the agent (each “+” in Figure 1D represents the identification of one key). For LLMs in the same series (i.e., Claude or OpenAI), it was clear that more advanced models have had much better success than earlier or simpler versions in completing the hypothesis-generation and testing steps, generating a predicted list, and identifying the two keys.

In addition to the nine known pathogenic TNRs, the B-catalog lists eight 5'UTR TNRs as “associated” with diseases. These are either adjacent to known pathogenic loci or have been considered for disease analysis. The number of these putative pathogenic 5'UTR TNRs predicted by the agentic runs is included in parentheses in Figure 1D.

Putative hypotheses on what distinguishes the pathogenic repeats

For the five agentic runs that predicted both pathogenic TNRs, we examined the hypotheses they generated based on the features that distinguish pathogenic repeats from the background (Figure 2A). At a glance, it seems the output varies in style. For instance, the agentic run with Opus 4.5 T=0.7 categorized the hypotheses based on “BASELINE CHARACTERISTICS” and “ALPHAGENOME PREDICTION”. In comparison, other agentic runs did not. However, upon closer inspection, some findings were shared across different agentic runs, for instance, except (Gpt5.2 T=0.9), all agentic runs identified that higher variability seems to distinguish the pathogenic TNRs from background (highlighted in blue) (Figure 2A & Supplementary Figure 1A). Similarly, multiple agentic runs found that pathogenic repeats tend to have a greater impact on 3D contact maps than average repeats (highlighted in pink) (Figure 2A, Supplementary Figure 1B). This suggests that the expansion of pathogenic repeats has a greater impact on chromatin organization and may affect gene expression beyond their immediate loci.

Potential pathogenic 5'UTR trinucleotide repeats.

Interestingly, for the five agentic runs that identified both keys, the keys (i.e., known pathogenic repeats) all ranked in the top 20 of the 50-candidate list (Figure 2D). Given that there are over 6,000 5'UTR TNRs included in the analysis, the probability that both keys appeared in the top 20 by chance alone is less than 0.001 (hypergeometric test, $p = 0.00086$). Based on this observation, we focused on the top 20 candidates predicted by these 5 agentic runs and generated a consensus list from a summarized ranking (Figure 2D).

In addition to the keys, four other repeats were identified by all five agentic runs as potentially pathogenic (Figure 2D). One of those, X-149631735-149631780-CGC, resides in the 5'UTR of TMEM185A and has been associated with FRAXF⁸. It is actually one of the 8 5'UTR TNRs recorded in the B-catalog as “associated” with repeat-expansion diseases.

A literature search revealed that although the repeat (13-35476296-35476353-CTG) in the 5'UTR of MAB21L1 was not listed as pathogenic in the B-Catalog, it was identified to be unstable and potentially involved in neurologic disorders⁹. Similarly, the expansion of repeat (17-32142451-32142499-CCG) (RHOT1) was identified as associated with a blood trait in a recent study¹⁰. The top-ranked repeat (5-443220-443277-GGC) is located at the 5'UTR of *Exocyst Complex Component 3* (EXOC3). There is no report of expansion of this repeat, but the gene is associated with Joubert Syndrome¹¹, a neurological developmental disorder.

Consistency of agentic analysis

The inference process of LLMs is probabilistic in nature. To assess the consistency of the agentic analyses, we repeated the same 24 analyses in March 2026. At this time, the agentic run with Opus 4.5 identified both keys across all tested temperatures (Figure 2C, Supplementary Figure 2). Interestingly, during this run, Sonnet 4.5 identified both keys at all temperatures, even though none of the runs with Sonnet 4.5 did so in the early Dec_2025 run. For the GPT series, gpt5 T=0.9 identified both keys. However, two runs that identified both keys in December 2025, gpt5 T=0.5; gpt5.2 T=0.9, failed to identify any keys. The examination indicated that gpt5.2 (T=0.9) failed to complete the reasoning process. The gpt5 T=0.5 run completed the reasoning process and generated the top candidate list, but failed to identify any keys. This discrepancy of totally different outcomes with the same LLM and temperature highlights the stochastic nature of LLM inference. In light of this, the relative consistency of coding models (Claude Opus 4.5 and Sonnet 4.5) is impressive.

While the performance of certain LLM/Temperature combinations varied significantly, the top 12 consensus candidates from these two sets of runs overlapped by 75% (Figures 2B and 2C).

Discussions

Our exploration strongly suggested that the latest LLMs have great potential to assist in hypothesis-driven discovery and testing. As with other data analyses, the hypotheses are based on correlation and, on their own, do not provide causal information. Both the hypotheses and the role of the predicted pathogenic TNRs need to be experimentally

verified. But the fact that 5 agentic runs using the most advanced LLMs identified both keys in the top 20 of their predictions is impressive. It should be noted that our approach leveraged the synergy between human expertise and an AI agent's processing power. The initial inspiration, the selection of the dataset, and the design and optimization of the overall strategy remain largely human endeavors. However, the tasks of generating and testing a spectrum of hypotheses were beyond the authors' capacity and wouldn't have been possible without the AI agent.

We focused our analysis on trinucleotide 5'UTR repeats. This is due to both conceptual concerns and practical limitations. Conceptually, STRs in different gene regions and repeats with varying motif lengths may have distinct pathogenic mechanisms. Focusing on 5'UTRs and limiting the analysis to TNRs helps ensure uniformity among the positive examples. Practically, the AlphaGenome predictions of the expansion dynamic of the 6,650 5'UTR TNRs generated a dataset of over 120 GB. As such, the agentic process required 512GB of RAM. Interestingly, the Grok4 T=0.7 agentic run started by generating a Parquet database for the AlphaGenome predictions. This approach may be necessary to explore much larger datasets, such as repeats in introns or other regions.

The summary of GPT5.2 differs from the others in that it claims there were no solid hypotheses beyond a “weak” one (Figure 2A). Yet, it nonetheless predicted both keys. Examination of the reasoning process recorded in the .log and .md files revealed that, in this run, the agent mostly relied on coefficient values from the data recorded in the B-catalog, such as population variability data from the “LPSLengthStdevFromHPRC100” and “StdevFromT2TAssemblies” columns. This suggests that genomic impact prediction from AlphaGenome is not essential for identifying pathogenic repeats. However, given that the other agentic runs used AlphaGenome output to formulate hypotheses and select candidates, we consider including this data in future analyses as appropriate.

Our results also suggest that the combination of CodeAct-based agent (Biomni) and coding LLMs (Claude Opus 4.5) provided reasonable repeatability in this kind of hypothesis-generation endeavor (Figure 2). Not only did all of the Opus 4.5-based runs identify both keys in the March 2026 run, but the top 7 consensus candidates predicted in December 2025 were also all among the top 12 in the March 2026 runs (Figure 2C). Thus, it appears that while the outcome of LLM-assisted exploration may differ at the individual-run level, obtaining consensus across different LLMs and running conditions greatly mitigates this variation.

Methods

Preparation of STR Datasets (Broad Institute catalog)

The JSON file of the B-catalog (v1.0) was downloaded from (<https://trexplorer.broadinstitute.org/>). The repeat catalog was filtered to include those located in 5'UTRs and whose motifs contain three nucleotides, and the results were output to a .tsv file. This dataset was further filtered based on the canonical motif (CCG & AGC). The “FoundInKnownDiseaseAssociatedLoci” and “KnownDiseaseAssociatedLocus” columns were removed before the dataset was provided to the agent.

AlphaGenome inference of the impact of repeat expansion

For all of the TNRs in 5'UTR used for this study (6,650), the repeat loci defined by the B-catalog were extracted from the hg38 reference genome and used as REF in the input file. Random expansions of the repeats at 2, 5, and 20 times were used as ALT in their respective input files (_2x, _5x, _20x). Batch inferences were performed using the AlphaGenome API provided by DeepMind (<https://deepmind.google.com/science/alphagenome/>). Since the output files were over 40GB, they were further divided into “_exp” for gene expression, “_epi” for epigenetic outputs, and “_splice” for those related to RNA splicing.

Implementation of the Biomni AI Agent and adaptation to this project.

We installed a copy of the Biomni agent with the associated data lake according to the standard procedure outlined by the authors¹². The codebase was slightly modified so it can work with all major LLMs and was renamed as Biomni-Rpts_Ds.

A list containing 7 of the 9 5'UTR pathogenic TNRs (red in Figure 1A) was provided to the agent. Two pathogenic repeats were withheld from the agent and used as keys for evaluation (Figure 1C).

For each agentic run, the agent is instructed to first identify what distinguishes the 7 pathogenic TNRs from the other >6,000 TNRs in 5'UTRs. The agent is specifically instructed that there is no published literature on this topic so do not seek any literature. Rather, the agent is commanded to develop novel hypotheses and test them using either information provided in the data files (above) or the databases compiled in the data lake for the Biomni agent5, including GTEx and MSigDB. Based on the mechanistic findings from the first stage, the agent is requested to develop a scoring mechanism to predict additional 5'UTR TNRs that might be pathogenic when expanded and report the top 50 candidates in a .csv file. (Figure 1C) (actual prompt file in Supplementary file 4).

- 1.) First, identify what distinguishes the 7 pathogenic repeats from the other about 6,000 repeats. The agent is specifically instructed that there is no published

literature on this topic, so do not seek any literature but rather develop novel hypotheses and test those with either information provided in the data files (above) or the databases compiled in the data lake for the Biomni agent, which include databases such as GTEx and MSigDB. The agent is requested to record the entire reasoning process and generate a .ipynb file for all hypotheses identified as significant.

- 2.) Based on the mechanistic findings from stage 1, the agent is requested to develop a scoring mechanism to predict additional 5'UTR repeats that might be pathogenic when expanded and report the top 50 candidates in a .csv file. (Figure 1C) (actual prompt file in Supplementary file 4).

Evaluation of LLM models

We primarily evaluated each model's performance based on whether it identified one or both of the known pathogenic repeats that were withheld from the agent. In addition, there are eight 5'UTR repeats listed as disease-associated in the B-catalog. These were considered either adjacent to known pathogenic loci or had been historically considered for rare disease analysis¹. We also recorded how many of these were among the top 50 predicted by the agents (numbers in parentheses in Figure 1D).

For each run, the agent was instructed to: (1) generate an .md with a detailed record for each step; (2) for each hypothesis on what distinguishes the pathogenic repeats, provide supporting statistical analysis and visual presentation in a .ipynb file; (3) predict the top 50 pathogenic candidates based on the hypotheses discovered and verified; (4) generate an .ipynb document on how the top 50 candidates were identified. These files were examined following the run to verify: (1) the agent did not do a literature search to include the two keys in the prediction; (2) the identification process was repeatable. The .ipynb generated by the gpt5.2 T=0.9 run was abbreviated, and additional details had to be ascertained from the .md and .log files)

Comparison of different LLMs and decoding temperatures

LLM and decoding temperatures were specified each time when the agent was initialized. The following LLMs were tested in both the December 2025 and March 2026 runs. API keys were provided in the .env file. "s3.5": "claude-3-5-sonnet-20241022"; "s4.5": "claude-sonnet-4-5-20250929"; "o4.5": "claude-opus-4-5-20251101"; "gem3p": "gemini-3-pro-preview"; "grk4": "grok-4"; "gpt4o": "gpt-4o"; "gpt5": "gpt-5"; "gpt5.2": "gpt-5.2". Additional LLMs were tested during preliminary testing, including "s3.7": "claude-3-7-sonnet-20250219"; "s4": "claude-4-sonnet-latest"; "gem2": "gemini-2.0-flash-exp"; "gem3f": "gemini-3-flash-preview".

The following additional LLMs were tested in March 2026: "s4.6": "claude-sonnet-4-6"; "o4.6": "claude-opus-4-6"; "gpt5.4": "gpt-5.4".

Generation of the consensus candidate list.

For the agentic runs that identified both keys. The top 20 candidates were used to generate the consensus list. For each repeat, the cumulative score is calculated with the formula $\text{score} = 20 - \text{Rank_in_the_run}$. The ranking is then sorted high-low based on their cumulative scores.

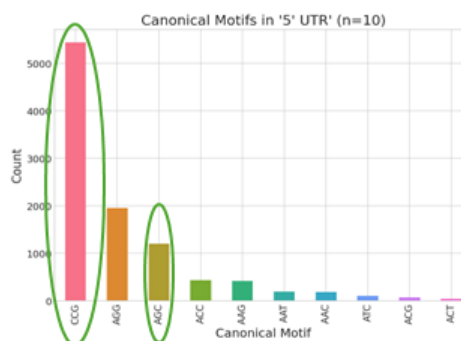
References:

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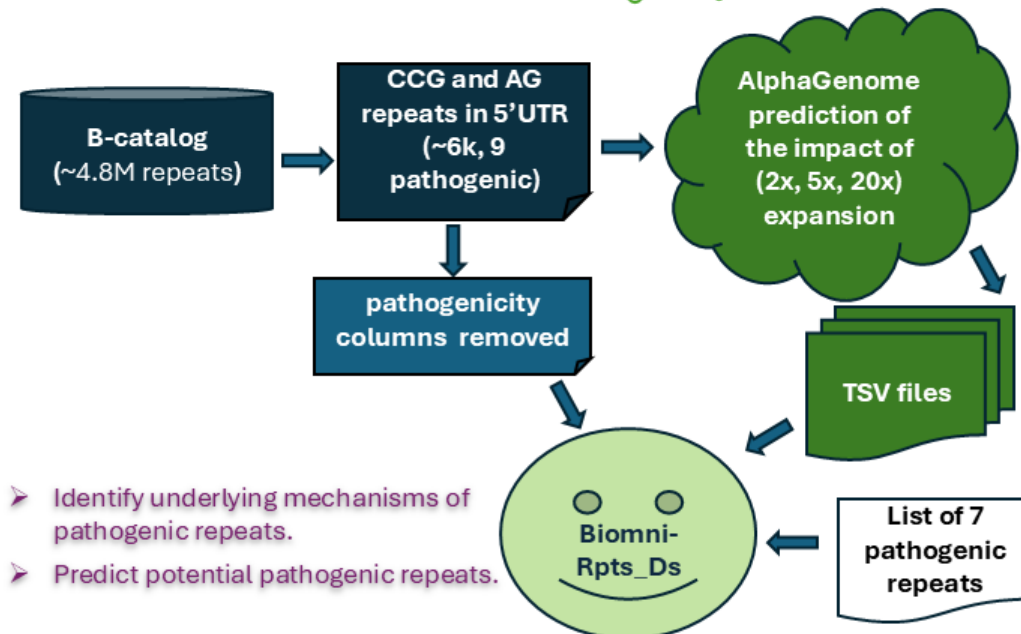
A

LocusId	Canonical Motif	KnownDisease Locus
1-94418421-94418442-GCC	CCG	ABCD3
X-148500631-148500691-GCC	CCG	AFF2*
12-50505001-50505022-GGC	CCG	DIP2B
X-147912050-147912110-CGG	CCG	FMR1
19-14496041-14496074-CCG	CCG	GIPC1
2-190880872-190880920-GCA	AGC	GLS*
8-104588970-104588997-CGC	CCG	LRP12
5-146878727-146878757-GCT	AGC	PPP2R2B
16-17470907-17470922-GCC	CCG	XYLT1

B



C



D

	LLMs	Temperature		
		0.5	0.7	0.9
Claude	Opus 4.5	++ (4)	++ (6)	++ (5)
	Sonnet 4.5	+ (5)	- (4)	X
	Sonnet 3.5	X	X	X
XAI	Grok 4	X	- (0)	X
Google	Gemni 3 pro	X	X	- (1)
OpenAI	GPT5.2	+ (3)	+ (4)	++ (4)
	GPT5	++ (6)	- (0)	X
	GPT4o	- (0)	- (0)	X

“+” – one known pathogenic repeats predicted (total 2).

(#) – number of disease-associated repeats predicted (total 7).

“X” – failed to generate prediction list.

Figure 1. **Overview of the strategy and comparison of LLMs for the agentic run.** **A.** List of 9 pathogenic TNRs in the 5'UTR. The seven in red were informed to the agent as positive examples, the two in blue were withheld from the agent and used as keys to evaluate the agentic runs. **B.** Histogram of 5'UTR TNRs in the B-catalog based on the canonical motif. **C.** The overall strategy of the agentic run. **D.** Comparison of agentic runs with different LLMs and decoding temperatures

A	O4.5 (T=0.5) (56 steps)	O4.5 (T=0.7) (49 steps)	O4.5 (T=0.9) (56 steps)	gpt5 (T=0.5) (7 steps)	gpt52 (T=0.9) (15 steps)
	Feature 1: Pathogenic rpts show significantly higher chromatin contact changes upon expansion. Feature 2: Pathogenic rpts show steeper dose-response relationships. Feature 3: Pathogenic repeats have longer repeat tracts in the reference genome. Feature 4: Pathogenic rpts show much higher length variability in the population.	1. BASELINE CHARACTERISTICS a) Higher Number of Repeats in Reference *($p < 0.001$, effect size = 0.93); b) Higher Population Variability - LPS Length Stdev; c) Higher Variability in T2T Assemblies. 2. ALPHAGENOMIC PREDICTIONS (Significant FDR) : a) Stronger 3D Chromatin Effects at 20x Expansion b) More Balanced Chromatin Accessibility Effects c) More Balanced TF Binding Effects; d) Higher Dose-Response in Chromatin Contact Effects.	1. NUMBER OF REPEATS IN REFERENCE *($p=3.29e-07$) - Pathogenic: mean=9.86 repeats, Non-pathogenic: mean=3.73 repeats. 2. POPULATION VARIABILITY. 3. T2T ASSEMBLY VARIABILITY 4. DOSE-RESPONSE SPLICE EFFECTS	1) Directional consistency in epigenomic tracks at 20x (and already detectable at 2x/5x) is significantly higher in pathogenic loci. 2) Pathogenic loci show substantially higher variability of the longest pure segment in HPRC long-read data, consistent with predisposition to repeat-length changes and instability. 3) Signals of epigenomic extremes in brain biosamples are relatively enriched among pathogenic loci at low expansion levels	Overall conclusion- There is no single robust univariate AlphaGenome consequence signature that cleanly separates the 7 pathogenic loci from all others across 2x/5x/20x. The best-supported hypothesis from this run is a weak neuronal/brain-skewed effect tendency, which is consistent with many known repeat-expansion diseases being neurologic, but it remains hypothesis-generating rather than "validated".

B

Top-12 Consensus Candidates — Dec_25

Rank	Gene	Rpts_LocusId	o4.5 T=0.5	o4.5 T=0.7	o4.5 T=0.9	gpt5 T=0.5	gpt52 T=0.9
1	EXOC3	5-443228-443277-GGC	4	1	2	2	3
2	GLS	2-190808072-19080920-GCA	3	5	4	6	1
3	RHOT1	17-32142451-32142499-CCG	9	3	1	8	5
4	MAB21L1	13-35476296-35476353-CTG	13	2	3	5	4
5	TMEM185A	X-149631735-149631780-CGC	8	4	6	7	7
6	AFF2	X-148500631-148500691-GCC	1	15	11	11	6
7	CARM1	19-10871583-10871649-GCG		6	7	16	2
8	BCLAF3	X-19990922-19990973-CCG	5	9	13		8
9	LINGO3	19-2388142-2388175-GCC		7	8	3	
10	FAM193B	5-177554489-177554531-CGC		14	12	4	14
11	BCL2L11	2-111120967-111121021-GCC	10	10	14	20	10
12	ENSG00000249773	7-55887686-55887639-GCG		12	5	13	15

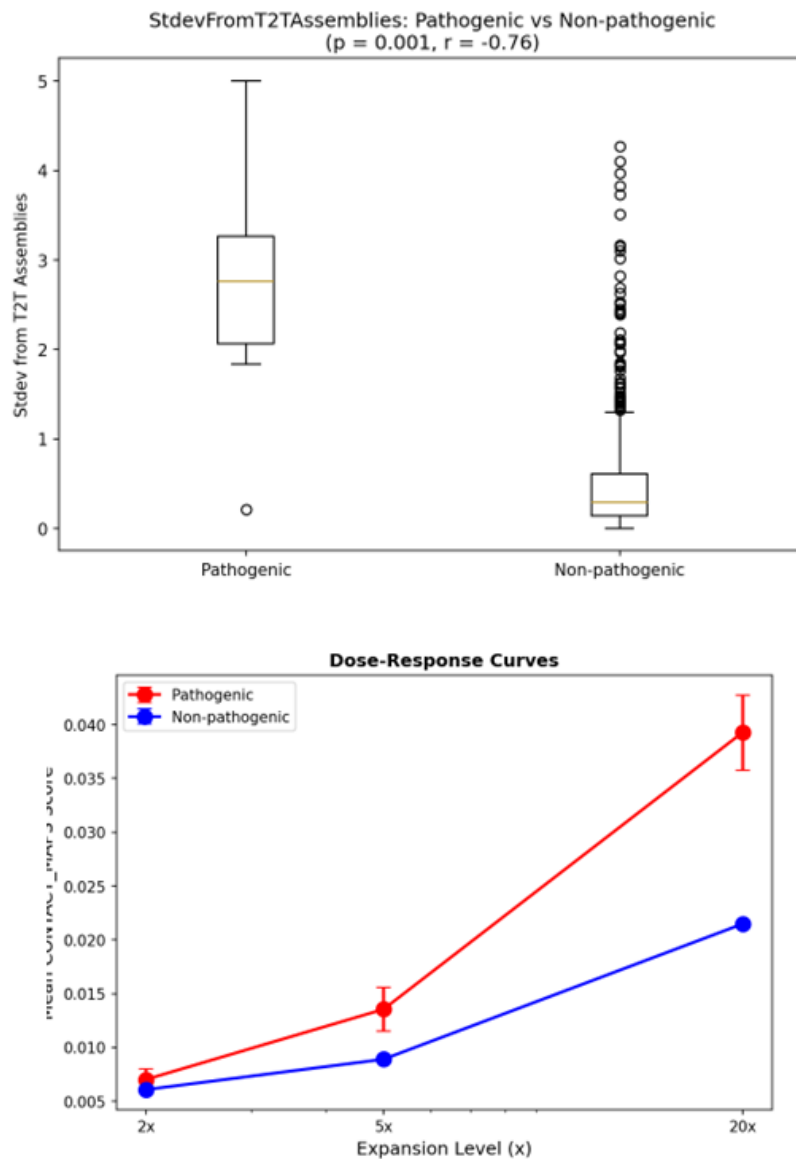
C

Top-12 Consensus Candidates — March_26

Rank	Gene	Rpts_LocusId	o4.5 T=0.5	o4.5 T=0.7	o4.5 T=0.9	gpt5 T=0.5	o4.5 T=0.5	o4.5 T=0.7	o4.5 T=0.9
1	EXOC3	5-443220-443277-GGC	1	2	2	5	1	1	5
2	MAB21L1	13-35476296-35476353-CTG	4	4	1	4	2	2	1
3	GLS	2-190808072-19080920-GCA	8	5	3	12	4	4	9
4	AFF2	X-148500631-148500691-GCC	2	3	11	3	14	10	3
5	CARM1	19-10871583-10871649-GCG	3	1	7	1	7	9	
6	UMAD1	7-7673310-7673378-AGC	5	7	8	2	9	12	14
7	TMEM185A	X-149631735-149631780-CGC	11	10	5	17	5	6	7
8	RHOT1	17-32142451-32142499-CCG	7		4	16	3	5	
9	LINGO3	19-2388142-2388175-GCC	20		6		6	3	6
10	BCL2L11	2-111120967-111121021-GCC	6	6	13	8	16	15	19
11	HMG3	X-150983398-150983452-CCG	9	8	17	7	10	19	
12	TENM2	5-166979194-166979236-CAG	13	16	12		11	13	8

Figure 2. Output from the AI Agent. **A.** Excerpts of the summaries generated by the agentic runs on “what distinguishes the pathogenic repeats”. Underneath the LLM and temperature is the number of steps it took the Biomni_Rpts_Ds agent to finish the process. Similar findings across these runs were highlighted. (* Similar statistical analyses were generated for all hypotheses. The following ones were omitted for simplicity; refer to the supplementary file for details). **B & C** Output plot generated by the agent. **B.** Top 12 consensus pathogenic repeats based on the December 2025 run. The highlighted numbers indicate their ranking in the respective agentic run. Empty space means it was not in the top 20 of predictions. The two known keys are enclosed in red boxes. **C.** Top 12 consensus candidates based on the March 2026 run. The repeats that were also identified in the December 2025 run were underlined with green lines.

Supplementary Figure 1.



Supplementary Figure 1. Output plot generated by the agentic runs. A. Pathogenic repeats have higher variability in T2T assemblies (O4.5, $T=0.7$). **B.** Pathogenic repeats show higher chromatin contact changes upon expansion (O4.5, $t=0.5$).

Supplementary Figure 2.

		Temperatures		
	LLMs	0.5	0.7	0.9
Claude	Opus 4.5	++ (6)	++ (5)	++ (7)
	Sonnet 4.5	++ (7)	++ (7)	++ (7)
	Sonnet 3.5	X	X	X
XAI	Grok 4	X	X	X
Google	Gemini 3 pro	X	X	X
OpenAI	GPT5.2	X	-(0)	X
	GPT5	-(0)	X	++ (5)
	GPT4o	X	-(0)	-(3)

Supplementary Figure 2. **Summary of the outcome of the March 2026 agentic run.**

Supplementary dataset.

1. List of pathogenic repeats in 5' UTR.
2. Table of all 5'UTR TNRs.
3. Query_05.txt
4. Examples of output from agentic runs (.zip)